

**Performance relative to the persistence baseline for WEATHER**  
**Lookback = 24 steps, Forecast horizon = 24 steps**

| Model         | RMSE / persistence | MAE / persistence | RMSE gain    | BASE_RMSE | BASE_MAE |
|---------------|--------------------|-------------------|--------------|-----------|----------|
| CNN           | 0.298              | 0.302             | 0.702        | 125.043   | 38.395   |
| LSTM          | 0.293              | 0.271             | 0.707        | 125.043   | 38.395   |
| dCeNN+ELM     | 0.299              | 0.316             | 0.701        | 125.175   | 38.448   |
| dCeNN+ELM+ASP | <b>0.291</b>       | <b>0.268</b>      | <b>0.709</b> | 125.175   | 38.448   |

- These ratios compare each model with the persistence baseline. Values below 1.0 for RMSE/MAE ratios mean the model beats persistence.
- Persistence is a simple forecast that repeats the last observed value forward.
- Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.